BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, DI KHAN

INTERMEDIATE (12TH CLASS) EXAMINATION, 2022

CHEMISTRY

Past Paper 2022 • Paper Code: DIK-HSSC-XII-2022 (A)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION – A

Time Allowed: 20 min | **Maximum Marks:** 20 | **Paper Code:** DIK-HSSC-XII-2022 (A)

Q1. Choose the correct option i.e. A,B,C,D and fill it up in the correct bubble in the given MCQs sheet.

Qi. Lassaign's test is used for the identification of _____ in organic compounds.

(A) Carbon
(B) Sulphur
(C) Hydrogen
(D) Oxygen

Qii. Acetylene can be regenerated from acetylides by treating it with _____

(A) H_2CO_3
(B) HNO_3
(C) H_2PO_3
(D) $HCLO_4$

Qiii. How many isomers are possible C_2H_6O?

(A) 4
(B) 8
(C) 2
(D) 5

Qiv. Which one of the following Alkyl halide is more reactive towards nucleophile substitution reaction?

(A) C_2H_5Br
(B) C_2H_5I
(C) C_2H_5F
(D) CH_3Cl

Qv. Geometry of methyl chloride is_____

(A) Planar
(B) Octahedral
(C) Tetrahedral
(D) Linear

Qvi. Which of the following alcohol don't undergoes oxidation?

(A) Tertiary Alcohol
(B) Primary Alcohol
(C) Secondary Alcohol
(D) Both A & B

Qvii. Williamsons synthesis is used for the preparation of _____

(A) Ketones
(B) Ethers
(C) Aldehydes
(D) Carboxylic acids

Qviii. Oxidation of alkyl benzene produced _____

(A) Benzoic acid
(B) Phenol
(C) Benzene
(D) Toluene

Qix. The Glyosidic linkage is present between the molecules of:

(A) Fatty acids
(B) Glucose
(C) Glycine
(D) None of these

Qx. Which one of the following is example of addition polymerization?

(A) Polyvinyl chloride
(B) Polyesters
(C) Polyamides
(D) Nylon 6,6

Qxi. Ozone layer in upper atmosphere is being destroyed by:

(A) CO_2
(B) SO_2
(C) Smog
(D) Chlorofluorocarbons

Qxii. Sigma to sigma star transition occurs in _____

(A) Alkane
(B) Ethene
(C) Ethyl alcohol
(D) Methyl chloride

Qxiii. Which of the following have high melting point?

(A) S
(B) P
(C) Cl
(D) Ar

Qxiv. Which of the following oxide is amphoteric nature?

(A) Na_2O_2
(B) Al_2O_3
(C) SiO_2
(D) SO_3

Qxv. Which of the following alkali metal only form normal oxide with O_2?

(A) Na
(B) K
(C) Li
(D) Cs

Qxvi. Brass is an alloy of _____

(A) Cu + Zn
(B) Fe + Cu
(C) Cu + Sn
(D) Mn + Ni

Qxvii. The Bidentate ligand among the following is:

(A) Co
(B) H_2O
(C) NH_3
(D) Ethylenediamine

Qxviii. The first organic compound prepared in laboratory was:

(A) Peat
(B) Sugar
(C) Urea
(D) Propanol

SECTION – B

Time Allowed: 2:40 Hrs | **Maximum Marks:** 40

Q2. Attempt any Ten questions of the following. All questions carry equal marks

i. Write the chemical reaction of water with oxides of Group 1A elements.

ii. Write the trend in solubility of hydroxides and sulphates of Group IIA elements.

iii. Write the systematic names of $Fe(CO)_5$, $[Cr(NH_3)_3(H_2O\}_3]Cl_3$, $[Na_2(NiCl_4)]$, and $K_4[Fe(CN)_6]$

iv. Write down the reactions of Hexaaquairon (II) and Hexaaquairon (III) with water.

v. How coal is converted to liquid hydrocarbons (Petroleum) by Fischer Troph (FT) method?

vi. Write at least four physical properties of alkanes.

vii. What is meant by Homolysis and Hetrolysis? Explain your answer with the help of example.

viii. Write any two methods for the preparation of amines.

ix. Prepare ethyl alcohols from the reduction of carboxylic acids and Esters.

x. Prepare aldehyde from ozonolysis of alkenes and hydration of alkynes.

xi. Why methanoic acid (formic acid) is stronger acid than ethanoic acid (acetic acid)?

xii. Why vegetable oils are sensitive to rancidity?

xiii. State and explain Bear-Lambert law.

SECTION – C

Maximum Marks: 27

> Note: Note: Attempt any three Question. Each question carries equal marks.

Q.3 a. Explain the trends in ionization energy, electronegativity, electrical conductivity and melting point of 3rd period elements.

b. How Beryllium (Be) differs from other member of its group?

Q.4 a. Discuss the chromate (CrO_4^{-2})-dichromate ($Cr_2O_7^{-2}$) equilibrium.

b. Starting from an alkene how will you prepare the following compounds? i. Alkane ii. Vicinal dihalides iii. Alkyl halide iv. Epoxide v. Alcohol

Q.5 a. Write the systematic names of the following organic compounds. [structural diagrams i–v in source scan — not transcribable as text]

b. Draw the structural formula of the following organic compound. i. 3-ethyl-4-methyloctane ii. Cis-hex-3-ene iii. 4-isopropyldecane iv. 2,4,4-trimethylhex-2-ene

Q.6 a. What is meant by water pollution? What are the main sources of water pollution?

b. What is meant by cracking? Discuss the different types of cracking in detail.